

Statistical Modeling: The Two Cultures

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¹ Leo Breiman (2001) | STAT 405

Context

Starting point

- **Leo Breiman:** Berkeley statistician; former consultant
- Worked on EPA, court, transportation, and prediction projects
- Main habit: solve the data problem, not defend a favorite model
- Validate on future or held-out data whenever possible

New problems

Bigger datasets, more compute, and more applied prediction tasks made simple prespecified models strain:

- many variables
- nonlinear structure
- hidden interactions
- weak signals spread across features

Motivating examples

LA ozone: EPA wanted 12-hour advance warnings. Breiman’s data had hundreds of meteorological variables and lags. The task was prediction-first: build something that works on future days.

Mass spectra: chlorine detection worked naturally as chemistry-shaped yes/no questions in a decision tree, not as a fixed-form discriminant model.

Status quo

- Default workflow: choose a stochastic model; estimate and report coefficients/tests/intervals
- If assumptions are weakly checked, model outputs can still travel as “significant” scientific claims
- Model conclusions can travel farther than the assumptions that justify them

Arguments

The split

Data modeling

- Assume a model for the black box
- Explain through parameters
- Rely on diagnostics and inference

Algorithmic modeling

- Treat the black box as unknown
- Build a predictive function
- Rely on held-out performance

Main claim

Match the modeling culture to the problem. When predictive power is prioritized, algorithmic approaches should take precedence.

Technical failures

- Goodness-of-fit and residual checks lose power in high dimensions
- Model tinkering after seeing data weakens classical tests
- Subset selection can make inference unstable
- Different plausible models can imply different scientific stories

Stress tests

Breiman’s razors cut both ways:

- **Multiplicity:** many good models can tell different stories; ensembles answer by averaging/voting across models
- **Simplicity:** readable models can underfit; complex models can predict better but need explanation
- **Dimension:** deleting variables can discard weak signal; algorithms can combine many small signals

Takeaways

What changed

Modern data-science practice echoes Breiman’s argument:

- out-of-sample validation as model comparison
- algorithms and ensembles in the standard toolkit
- interpretations beyond coefficients
- problem-first choice of predictive, inferential, or causal tools

Black boxes

Breiman’s claim is not that black boxes are magic. It is that accurate predictive models can still provide valid information:

- variable importance
- interactions and subgroups
- stable predictive structure across tests

That information still needs domain judgment and validation.

Questions

- What assumptions replace data-model assumptions?
- What makes black-box information valid enough to act on?
- When do prediction, inference, causality, and interpretability need different tools?
- What mix of data models and algorithms does a problem require?

Reference

Breiman, Leo. 2001. “Statistical Modeling: The Two Cultures.” *Statistical Science* 16 (3): 199-231.

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